Abhishek Kumar Jha

Senior Software Engineer | Backend & AI Infrastructure | Python · Distributed Systems · APIs

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience building scalable backend systems, distributed platforms, developer-facing APIs, and production AI infrastructure. Hands-on experience designing agent orchestration runtimes, AI guardrails, observability, and cloud-native systems using Python/FastAPI, Java/Spring Boot, PostgreSQL, AWS, Docker, and Kubernetes. Experienced taking complex systems from architecture through deployment, testing, monitoring, and production operation.

TECHNICAL SKILLS

Backend & Platform: Python, FastAPI, Java, Spring Boot, TypeScript, Node.js, REST APIs, GraphQL, Distributed Systems, Microservices, Workflow Orchestration

AI Infrastructure: LLMs, GenAI, AI Agents, Agentic Workflows, Tool Calling, RAG, Model Routing, AI Guardrails, Retrieval

Cloud & Infrastructure: AWS, ECS Fargate, ALB, ECR, RDS, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD

Data: PostgreSQL, SQL, SQLAlchemy, Alembic, Data Integrations

Reliability & Quality: Pytest, Vitest, Playwright, Automated Testing, Observability, Monitoring, Splunk, Root-Cause Analysis, Production Troubleshooting

PROFESSIONAL EXPERIENCE

Sudo Programmer Inc. — *Founder & Senior Software / AI Platform Engineer (Concurrent Venture)* 2025 – Present

- **Agentic SDLC Platform:** architected and built a governed AI platform (Python/FastAPI, PostgreSQL, AWS) that orchestrates multi-step agent workflows from requirements through code generation, testing, and GitHub pull requests, with human approval gates and end-to-end traceability.
- Designed a fault-tolerant distributed execution runtime with scheduling, lease-based workers, DAG orchestration, checkpoint recovery, and pause/resume/replay/fork capabilities, separating deterministic execution from LLM-driven decision making.
- Built guardrails and governance controls for AI-generated changes, including scoped file access, secret detection, architecture-contract enforcement, model routing, retry policies, per-run cost limits, and operational telemetry.
- Delivered ~190 REST APIs and 700+ automated tests across Pytest, Vitest, and Playwright; shipped containerized services through GitHub Actions to AWS ECS Fargate, ALB, and RDS.
- **PlanCraft AI:** built and launched an AI productivity platform using Python, FastAPI, PostgreSQL, AWS, LLMs, RAG, tool calling, and agentic workflows, serving 1,000+ users.

CACI International, Remote — *Senior Java Full Stack Developer* Jul 2025 – Present

- Design and operate Java/Spring Boot microservices and REST APIs for distributed production systems, contributing to API architecture, security, deployment, and reliability.
- Containerize services with Docker, deploy through Kubernetes environments, and maintain CI/CD workflows using Jenkins and Git, with security via Spring Security, OAuth 2.0, and JWT.
- Diagnose production incidents using application logs and Splunk monitoring data, performing root-cause analysis and reliability improvements.

Braintree Health, Corpus Christi, TX — *Software Engineer Intern / Team Lead* Nov 2024 – Jun 2025

- Built Java/Spring Boot REST API services for complex healthcare data integrations (HL7, FHIR, DICOM) and resolved production data-processing issues, reducing a key workflow from 2–3 weeks to ~1.5 hours.

Eruvaka Technology, India — *Java Full Stack Developer / Team Lead* Nov 2021 – Aug 2024

- Led and mentored engineers while designing backend services and APIs (TypeScript, Node.js, GraphQL, REST) for a distributed platform serving 20,000+ users.
- Collaborated with backend and data-science teams to integrate ML-driven capabilities, including growth prediction and harvest optimization, into production systems.

Chubb Insurance / Accenture, India — *Java Full Stack Developer / Software Engineer* Jul 2019 – Nov 2021

- Designed and delivered enterprise Java backend services and APIs, reducing page-load times from ~10–12 seconds to 3–4 seconds and delivering a major platform approximately one month ahead of schedule.

ADDITIONAL EXPERIENCE

Xenovex Technologies — Java Developer (Jan 2018 – Jun 2019): Built enterprise Java applications, backend services, and API integrations across the software development lifecycle.

EDUCATION & CERTIFICATIONS

M.S. Computer Science, Texas A&M University–Corpus Christi · B.Tech Computer Software Engineering, SRM Institute of Science & Technology

AWS Certified Cloud Practitioner · Databricks Fundamentals · 2× Best Performer Award — Accenture